

forecast from the Survey of Professional Forecasters predicted current CPI headline inflation during that quarter to be -2.6% , coming back to 2.0% during 2009. Thus the negative break-even inflation rates over 2008 and 2009 were consistent with market participants forecasting dire, Japan-like deflation scenarios during that time. At the same time, the flight to safety in Treasuries, which forced up Treasury prices and drove down Treasury yields, was consistent with Treasuries having a negative risk premium. Investors so desperately wanted Treasuries that they were willing, on a risk-adjusted basis, to pay to hold them (there was a negative risk premium) rather than demanding to be paid to hold them (the normal positive risk premium). So both the deflation scenario and the negative risk premiums were important.

4.5. ILLIQUIDITY

An important factor in TIPS markets is illiquidity risk. The TIPS market was illiquid during the first few years after TIPS were introduced, up to around the early mid-2000s, and is still much less liquid than Treasuries.¹⁴ Since many of the investors preferring TIPS are pension funds and retail investors, Sack and Elasser (2004) predict the TIPS market will remain more illiquid than the Treasuries market, where there is a larger concentration of active fund managers.

Researchers estimate the TIPS illiquidity risk premium was around 1.0% for the five-year TIPS and between 0.5% and 1.0% for ten-year TIPS up until around 2003 and 2004. In 2004, illiquidity risk premiums fell substantially to well below 0.5% for both the five- and ten-year TIPS. They dramatically spiked to above 2.5% during the financial crisis, and after 2009 have gone back to below 0.5% .

The illiquidity risk premium makes TIPS yields higher than what they would be if they had the same liquidity as Treasuries. That is, liquidity-adjusted real yields would be *lower*, and thus even more negative at December 2011, than what we observe in TIPS markets. Liquidity-adjusted inflation compensations would be also *higher* than the raw inflation compensations. Thus if negative real rates reflect investors' expectations of very bad times in the future, the bad times are predicted to be even worse after adjusting for liquidity risk.

4.6. TIPS AND TAXES

Interest payments on TIPS, like the interest paid on conventional Treasury bonds, are subject to Federal income tax. Any inflation-adjusting increases in the principal of TIPS are also taxable. The tax code does not differentiate between real and nominal income—it is just income—so the investor's after-tax yield is exposed to inflation risk. During periods of high inflation, the tax levied on the increased

¹⁴ See D'Amico, Kim, and Wei (2009), Gürkaynak, Sack, and Wright (2010), and Pflueger and Viceira (2011).